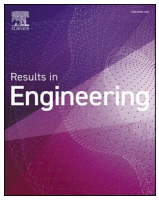
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Review article



Revolutionizing urban solid waste management with AI and IoT: A review of smart solutions for waste collection, sorting, and recycling

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ABSTRACT

This review explores the transformative impact of artificial intelligence (AI) and the Internet of Things (IoT) on urban solid waste management, focusing on their potential to enhance the processes of waste collection, sorting, and recycling. With urban populations on the rise and waste generation rapidly increasing, cities face critical challenges to environmental sustainability and public health. The integration of IoT-based smart waste management enables municipalities to utilize real-time data to analyze waste flow, predict generation trends, and improve operational efficiency. AI-powered sorting systems and advanced robotics can further optimize recycling rates by reducing the need for manual labor and increasing material recovery in recycling facilities.

Additionally, the necessity to leverage strong collaboration among governments, technology providers, and the public is emphasized in this paper as a means to develop effective policy frameworks and foster community engagement in sustainable waste practices. The review also addresses challenges such as high initial costs, privacy concerns, and technical limitations, advocating for continued investment in innovative technologies and robust data privacy solutions. Ultimately, this research promotes a holistic approach to sustainable waste management that centers on building a circular economy and reducing environmental impacts. By adopting AI and I oT advanced technologies, cities can achieve their sustainability objectives, thereby paving the way for cleaner, more resilient urban environments that will benefit both current and future generations.

1. Introduction

Industrialization and population growth have resulted in unprecedented levels of solid waste production worldwide [1,25,26,28,51]. Urban areas in particular have faced mounting environmental and public health challenges in managing the vast amounts of solid waste generated by the densification of population centers [11,27,28,45]. According to a recent report released by the World Bank, cities generate over 2 billion tons of municipal solid waste (MSW) annually, which is projected to increase to 3.4 billion tons by 2050 [48]. This surge in waste generation, coupled with limited landfill space and the environmental consequences of improper disposal, has made MSW one of the most pressing issues of our time [23,31,34]. Traditional waste management systems that previously relied on periodic collection, manual sorting, and landfill disposal now struggle to keep pace with rapidly increasing waste volumes [30]. These systems typically employ fixed collection schedules and basic sorting, which frequently results in overflowing bins in high-traffic areas, unnecessary greenhouse gas emissions, and inefficient resource allocation [16,30,33]. In an era where cities are under pressure to reduce their carbon footprint and meet sustainability goals, the limitations of such systems are even more apparent. Inadequate and improper waste disposal practices have also been shown to contribute directly to public health crises, with overflowing landfills and open dumping leading to the spread of disease and the polluting of water sources [16,37].

As waste generation quickly outpaces the capacity of traditional management approaches, the need for innovation has become more and more urgent [16]. Research confirms that the environmental and societal impacts of ineffective waste management are severe [5,16,34], with landfills releasing large amounts of methane, a potent greenhouse gas that significantly contributes to global warming [16,24]. Methane is more effective even than carbon dioxide at trapping heat in the atmosphere, which means that minor increases in landfill-generated methane emissions can have major climate impacts of both regional and global scales [9,16]. Furthermore, runoff from landfill sites, especially those located near urban areas, can pollute rivers, lakes, and even groundwater [16]. In some parts of the world, open dumping and waste burning are common practices that release hazardous pollutants into the air,

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causing respiratory illnesses and exacerbating health issues in nearby communities [16,40]. As a result, the challenges of MSW management extend beyond environmental concerns to also encompass health, equity, and economic considerations [6,16,50].

In recent years, the development and rapid evolution of digital technologies, particularly artificial intelligence (AI) and the Internet of Things (IoT), have provided new solutions to address emerging challenges [8,14,15,39]. Today, these technologies are fundamentally reshaping how cities collect, sort, and recycle waste [41]. AI enables systems to make intelligent decisions based on data, resulting in improved accuracy and efficiency in route planning, waste sorting, and other aspects of MSW management [41,42]. For example, AI-driven algorithms can analyze complex datasets to optimize collection schedules, which minimizes the frequency of pickups while still ensuring that bins are emptied before they overflow and that underused bins are not unnecessarily emptied [42]. The optimization of resources not only improves the reliability of collection systems but also reduces fuel costs and emissions.

Similarly, IoT technology is revolutionizing MSW management by providing real-time monitoring capabilities that enable waste bins and collection vehicles to communicate their status, location, and capacity levels to central management systems. Sensors embedded in waste bins monitor fill levels, while GPS-equipped vehicles track routes and operational performance. These systems communicate using advanced protocols such as LoRaWAN and NB-IoT, ensuring seamless data transmission to centralized platforms. IoT-enabled platforms then analyze the data and formulate the most appropriate response, thus leading to reductions in unnecessary trips, fuel consumption, and operational costs.

When paired with AI, IoT becomes even more transformative. AI algorithms can analyze IoT-generated data not only to predict waste generation patterns within urban environments, but also to dynamically adjust collection schedules and optimize resource allocation. For example, ML models can forecast peak waste volumes during events or specific seasons, enabling cities to respond proactively. Additionally, AI-powered automated sorting systems can improve recycling efficiency by categorizing waste with high accuracy, significantly reducing landfill contributions. The integration of AI and IoT succeeds both in enhancing waste collection and sorting and in fostering a circular economy, as recyclables can be tracked throughout their lifecycle, ensuring greater transparency and sustainability. By leveraging AI and IoT, urban centers can adopt SMART (self-monitoring, analysis, and reporting technology) and adaptive MSW management solutions that address the growing challenges of solid waste generation. This synergy reduces environmental impacts, makes the best use of available and potential resources, and supports cities in achieving sustainability goals, marking a paradigm shift in waste management practices. Smart waste management solutions utilizing AI and IoT have already demonstrated their enormous potential in many cities worldwide [2]. For example, smart bins equipped with IoT sensors can detect when they are nearly full and automatically signal collection services. This reduces the chance of overflow while also ensuring that collection resources are deployed only when necessary. Moreover, route optimization algorithms can analyze data on traffic, bin fill levels, and fuel usage to determine the most efficient collection routes and times. The elimination of unnecessary trips reduces fuel consumption and vehicle wear-and-tear, thus lowering the overall carbon footprint of waste collection activities. These improvements in efficiency are critical as cities seek to balance environmental responsibility with operational costs.

Another promising application of AI and IoT in waste management is automated sorting systems, which play a crucial role in recycling [47]. Traditional sorting facilities rely heavily on human labor, with workers manually separating waste based on visual cues. However, this approach is not only costly and time-consuming, costly, but it often also results in a significant amount of recyclable material being overlooked. AI-powered sorting systems can identify and categorize waste materials with high precision, even differentiating between various types of plastics, metals, and other recyclables. By automating the sorting process, AI significantly enhances the speed and accuracy of waste separation, which in turn increases recycling rates and reduces the amount of waste sent to landfills [47]. Beyond collection and sorting, AI and IoT can improve recycling processes by tracking and verifying materials throughout the end stage of their lifecycle. Blockchain technology, when combined with IoT, can create an immutable record of waste materials from the point of collection through to processing and recycling [21]. This level of traceability ensures that recyclables are not diverted to landfills and provides a high degree of transparency, allowing cities and companies to verify that recycling goals are being met. Transparency is a critical component in building consumer trust. Residents and stakeholders need to feel confident that their efforts to separate waste are contributing to meaningful environmental outcomes, or they may not participate in the programs. Moreover, AI-based platforms are increasingly being used to create incentives for recycling. Cities and private companies have started using data-driven rewards programs that incentivize residents to sort their waste correctly, fostering a culture of responsible waste disposal and supporting the transition to a circular economy.

The novelty of this review lies in its comprehensive exploration of the transformative potential of AI and IoT technologies specifically in the context of municipal solid waste management. While AI and IoT have been widely discussed in other industries, their applications in waste management have not. The transformative potential of AI and IoT in MSW management lies not only in their individual capabilities but in their synergistic integration, a perspective that remains underexplored in the current literature. This review introduces a novel framework that combines the complementary strengths of AI and IoT to create adaptive, efficient, and scalable waste management systems. By dynamically processing real-time data from IoT sensors with AI algorithms, this integrated approach optimizes decision-making processes that can be applied to numerous aspects of the systems, including route planning, predictive maintenance, and resource allocation.

The synergy that results from combining AI and IoT in MSW management allows for a more comprehensive response to emerging challenges. High population density, resource constraints, and environmental impacts require a sustainable and dynamic solution. In addition to exploring the potential of the integrated approach, this paper proposes a unique framework tailored to resource-constrained urban areas. Existing models in the literature cater predominantly to well-funded cities, but the framework presented here emphasizes modular designs and scalable architecture, enabling cities with limited budgets to gradually adopt smart waste management technologies over time, as their budget allows. The design addresses a critical gap in the literature while also expanding the applicability of AI and IoT innovations to diverse urban settings.

At the same time, this study provides a forward-looking perspective on the integration of blockchain technology with AI and IoT in waste management. Traceability and accountability are significantly enhanced by creating immutable records of waste material lifecycles. Furthermore, by leveraging smart contracts, this approach also offers innovative incentive mechanisms to encourage recycling behaviors, supporting the transition to a circular economy. These concepts succeed in advancing sustainability goals in large part because they address transparency and stakeholder trust, areas that are vital for the widespread adoption of the discussed technologies.

In synthesizing these innovative frameworks and examining their practical implications, the present work highlights actionable insights and provides a roadmap for future research, thus addressing both the technological and societal dimensions of MSW management. These contributions position AI and IoT as transformative tools capable of reshaping urban sustainability, public health, and community engagement. Moreover, by examining the unique challenges faced by waste management systems in urban environments (e.g., high population

density, resource constraints, and environmental impact), this article explains how AI and IoT can be tailored to address these issues in ways that traditional methods cannot. As well, this review synthesizes recent advancements in smart waste technologies, offering insights into both the practical applications of these systems and the potential barriers to their widespread adoption.

Finally, this article provides a foundation for understanding how digital innovations can revolutionize waste management and offers a roadmap for cities seeking to implement these technologies in pursuit of sustainability and operational efficiency. Case studies, comparative analysis, and a discussion of future research opportunities are presented that combine insights of technological advancements with a critical examination of the social, environmental, and regulatory factors influencing waste management practices. Ultimately, this review aims to provide a forward-looking perspective on how AI and IoT can address the multifaceted challenges of MSW management, underscoring the transformative potential of smart waste solutions in creating more sustainable, efficient, equitable, engaged, and resilient cities.

2. Technological advancements in waste collection

In recent years, advancements in AI and the IoT have begun to reshape MSW management, driving the development of smart technologies toward more efficient, sustainable, and environmentally friendly solutions [8,53]. Traditional approaches that rely heavily on routine collection schedules and manual sorting are being transformed by leveraging real-time data and intelligent decision-making processes. The integration of AI and IoT into waste management systems enables cities to handle the complex challenges of collection, sorting, and recycling with unprecedented efficiency and accuracy. Among the most impactful innovations are smart bins equipped with sensors, AI-driven route optimization algorithms, and predictive analysis models, which together offer a comprehensive approach to waste management in an increasingly urbanized world.

Smart bins are revolutionizing waste collection in cities worldwide [52]. Equipped with sensors that can measure waste levels, detect odors, and monitor bin status, IoT-enabled bins provide real-time data to MSW management systems. These data feed into a centralized system that monitors waste levels across a city, allowing for efficient scheduling and allocation of resources. For example, sensors within the smart bins can detect when a bin is nearing capacity and automatically alert collection services to dispatch a truck to empty it. By ensuring that only full bins are collected, waste management systems can avoid unnecessary trips, which not only reduces operational costs but also minimizes the environmental impact of collecting vehicles on the road. In some cases, smart bins are also capable of detecting odors. Odor detection helps prevent situations where a bin is nearly empty but the waste already contained within it has begun to rot or emit unpleasant smells in public areas. Being nearly empty, the bin might not trigger the signal for waste removal based on waste volume alone, but with the odor sensor on board, the appropriate signal is triggered. This feature is particularly valuable in high-traffic locations like city centers, parks, and tourist areas, where maintaining cleanliness is essential to both the local environment and public satisfaction. Real-time data provided by smart bins are crucial for optimizing resource allocation. In a traditional waste collection system, trucks follow fixed routes and schedules, which often means that bins are emptied before they are full, leading to inefficiencies in resource use and higher operational costs. In contrast, IoT-enabled bins allow for dynamic collection schedules that respond to actual waste levels. This approach significantly reduces the frequency of unnecessary collections, allowing waste management services to allocate their resources more efficiently. In cities where some areas produce more waste than others, smart bins can help direct collection services to focus more on high-usage zones while reducing attention on low-waste areas. Optimization is particularly beneficial where municipalities are facing budget constraints, as it reduces fuel consumption, vehicle wear-and-tear, and labor costs. Moreover, by minimizing the need for frequent trips, IoT-based smart bins significantly contribute to reducing greenhouse gas emissions, offering measurable environmental benefits. For instance, optimizing waste collection routes using real-time data from smart bins can reduce vehicle fuel consumption by up to 30 %. This reduction in fuel usage directly lowers carbon dioxide emissions, helping cities move closer to their environmental sustainability targets. In fact, cities implementing IoT-based systems for waste management can achieve up to a 25 % reduction in overall operational costs, largely driven by reduced fuel consumption and vehicle wear-and-tear. By optimizing routes based on real-time waste bin data, AI-driven systems can cut down on unnecessary trips, thereby reducing the total kilometers driven by waste collection vehicles. Smart waste management systems have led to a 20 % decrease in total driving distance, further contributing to a reduction in emissions. This highlights the tangible benefits of IoT and AI integration, demonstrating not just a qualitative shift in MSW management practices but also a measurable improvement in environmental performance, economic efficiency, and operational sustainability.

Along with fill-level and odor monitoring, smart bins can also be integrated with AI algorithms that analyze waste generation patterns over time. By collecting data on the types and quantities of waste deposited in different areas, AI-driven systems can develop insights into waste production habits across various demographics, locations, and timeframes. For example, bins located in commercial areas may experience higher levels of waste generation during business hours, whereas bins in residential neighborhoods may see peak usage in the evenings or weekends. AI systems can learn from these patterns and adjust collection schedules accordingly, ensuring that bins in high-usage areas are emptied more frequently during peak times. This adaptive approach enhances the efficiency of waste collection services by reducing instances of overflow in busy areas and minimizing underutilized collection efforts in low-usage zones. Route optimization is another significant advancement made possible by the combination of AI and IoT in MSW management [35]. AI algorithms can analyze a wide range of data inputs, including traffic patterns, bin fill levels, energy use, and vehicle capacity, to determine the most efficient routes for waste collection vehicles. Traditional waste collection routes often follow predetermined paths that may not reflect current road conditions or actual waste levels, resulting in inefficient routes that waste fuel and time. With AI-driven route optimization, waste management systems can update routes in real-time, accounting for factors such as traffic congestion, road closures, and high-waste areas that need immediate attention. By dynamically adjusting routes based on real-time data, AI-enabled systems help reduce the distance and time required for waste collection, thereby decreasing fuel consumption and greenhouse gas emissions. For instance, if certain routes experience heavy traffic during specific times of the day, AI can adjust schedules to avoid those periods, leading to smoother and more efficient operations. The environmental benefits of route optimization are considerable [35]. In addition, by reducing the number of trips and optimizing routes, cities can significantly lower their carbon footprint associated with waste collection. Emissions from waste collection trucks, which often rely on diesel fuel, contribute substantially to air pollution in urban areas. AI-driven route optimization not only cuts down on fuel use and emissions but also prolongs the lifespan of collection vehicles by minimizing wear-and-tear. This, in turn, reduces the frequency and cost of vehicle maintenance and replacement, providing both environmental and financial benefits for waste management systems. Moreover, the reduced presence of waste collection vehicles on the road alleviates traffic congestion, contributing to a smoother flow of traffic in busy urban areas. Cities implementing AI-driven route optimization have reported a noticeable decrease in operational costs and environmental impact, underscoring the transformative potential of AI in modernizing urban waste management [8,49]. Predictive analysis is yet another tool in AI's expanding toolbox. Powerful data processing capabilities allow cities to anticipate waste

generation patterns with remarkable accuracy. By analyzing historical data alongside variables such as seasonal trends, event schedules, and community demographics, AI systems can predict fluctuations in waste production and adjust collection schedules to reflect these changes. For example, during festivals, concerts, or major public events, waste generation often spikes, leading to a higher-than-average volume of waste in a short time frame. With predictive analysis, cities can proactively adjust their collection schedules to accommodate these temporary increases, ensuring that waste bins are not overfilled and that public spaces remain clean. This capability is especially valuable in cities with high tourism rates, where fluctuating population numbers can make waste management a complex and dynamic task. Predictive analysis helps prevent scenarios in which waste management services are caught off-guard by sudden surges in waste, leading to improved service reliability and better public satisfaction. AI-based predictive analysis can also consider long-term waste production trends, such as seasonal variations in waste generation. In many cities, certain times of the year, e.g., the holiday season or summer months, may experience higher waste levels due to increased consumption and tourism. By understanding these seasonal trends, MSW management systems can prepare in advance, deploying additional resources during peak times and scaling back during low-waste periods. Furthermore, AI algorithms can account for changes in community habits and policies, such as increased recycling rates or the introduction of waste reduction programs, allowing for ongoing adjustments to collection strategies. In this way, predictive analysis supports a more responsive and adaptive approach to waste management, helping cities meet their sustainability and efficiency goals while maintaining a high standard of public cleanliness. The integration of AI and IoT into waste management also opens up opportunities for enhanced recycling practices that contribute to the development of a circular economy [22,36]. Smart bins can be equipped with sensors that recognize the material composition of waste items, aiding in the separation of recyclables from general waste. By identifying materials such as plastics, metals, and glass at the point of disposal, smart bins can streamline the recycling process and reduce the volume of waste sent to landfills. AI-driven sorting systems within waste facilities further enhance this process by automating the identification and categorization of recyclables.

At the same time, the use of AI and IoT in recycling allows for more transparent tracking and verification of waste materials throughout the end stage of their lifecycle. Blockchain technology, in conjunction with IoT sensors, can create a record of waste from collection to processing, ensuring that recyclable materials are not diverted to landfills. This level of traceability enables cities to verify that their recycling goals are being met and provides stakeholders with the assurance that recyclable materials are being properly managed. Additionally, AI-powered platforms can incentivize recycling efforts by rewarding residents who properly sort their waste, fostering a culture of responsible disposal. Through these and similar data-driven rewards programs, cities can encourage sustainable waste practices and promote environmental stewardship among their communities.

The implementation of smart waste management technologies also aligns with broader sustainability goals. Cities worldwide are facing increased pressure to reduce their carbon footprints, and AI and IoT-based waste management systems provide a viable solution [20,35]. By reducing waste collection frequency, optimizing routes, and minimizing landfill usage through improved recycling, these technologies significantly contribute to a reduction in urban carbon emissions. Moreover, by diverting waste from landfills and enhancing recycling practices, smart waste management supports the transition to a circular economy, where resources are reused and repurposed rather than disposed of after a single use. This approach both conserves valuable resources and reduces the environmental impact associated with waste disposal.

3. AI and IoT in waste sorting

Automated waste-sorting systems are redefining the efficiency and accuracy of material recovery facilities (MRFs), particularly with the integration of AI and machine learning. In these facilities, waste is sorted, processed, and prepared for recycling, but traditional systems have often required extensive manual labor and involved high operational costs due to inefficiencies and limitations in sorting technology. However, with the use of AI-powered robotics, computer vision systems, and IoT-enabled smart sensors, the processing of waste has been revolutionized. These technologies are not only reducing the reliance on manual labor but also providing unprecedented accuracy in sorting, which plays a critical role in improving recycling rates and diverting waste from landfills. One of the key advancements in MRFs is the development of automated sorting systems that leverage AI to identify, classify, and separate different types of materials with remarkable precision. Robotic arms equipped with ML algorithms and sophisticated sensors are capable of recognizing various waste items, including metals, plastics, glass, paper, and even different types of plastics that traditional systems might struggle to distinguish. These advanced systems are trained on vast datasets that enable them to identify subtle differences in waste materials, ensuring that each item is sorted accurately and sent to the appropriate recycling stream. The automation of these tasks not only minimizes the need for human intervention but also increases sorting efficiency, allowing MRFs to process higher volumes of waste in shorter periods. At the heart of these automated sorting systems is the concept of machine learning, a subset of AI that allows robotic systems to “learn” from data and improve their performance over time. Through repetitive processing and exposure to different waste materials, AI-powered robots can refine their accuracy and adapt to new types of waste, such as emerging materials used in packaging. For example, if a new type of recyclable plastic enters the waste stream, ML algorithms can be updated to recognize its specific properties, thus ensuring that it is correctly sorted in the future. This adaptability is especially important as consumer products evolve and new types of recyclable materials are introduced into the market. AI-driven computer vision systems are a crucial component of automated sorting processes, as they enhance the speed and reliability of waste identification and categorization. By using computer vision, sorting facilities can scan conveyor belts in real-time, analyzing the shape, color, size, and texture of each item to determine its material composition. This enables highly efficient separation, as items are classified almost instantly and directed to the appropriate processing areas. For example, a piece of aluminum, regardless of its shape or size, can be quickly detected and separated from plastic, paper [44], and other materials. The speed at which computer vision systems operate allows for a continuous flow of materials through the sorting line, reducing bottlenecks and maximizing throughput. The accuracy of computer vision systems is particularly beneficial in separating complex items, such as mixed plastic packaging that may contain several layers of different materials. These systems can detect even subtle visual cues that indicate a specific type of plastic, allowing for more refined sorting. This, in turn, enables MRFs to meet the stringent quality requirements of recycling facilities that accept sorted materials for reprocessing. By ensuring that each material stream is as pure as possible, AI-enabled computer vision systems contribute to a more efficient recycling process, as materials can be processed with fewer contaminants and at lower energy costs.

The integration of AI and IoT in MRFs has birthed smart sorting systems that can adjust operations based on real-time data. IoT sensors embedded within sorting equipment continuously monitor waste composition, providing valuable insights into the types and quantities of materials being processed at any given time. This real-time feedback allows MRFs to dynamically adjust sorting priorities, focusing on materials that are in higher demand or adjusting sorting parameters when certain types of waste become more prevalent. For example, if a sudden increase in plastic waste is detected, the sorting line can be optimized to

prioritize plastics, ensuring that they are sorted quickly and effectively [7,32].

IoT-enabled sensors also provide MRF operators with information about the performance and wear of sorting equipment. Predictive maintenance helps prevent costly downtime by identifying potential issues before they lead to equipment failure. In this way, facilities can schedule maintenance during non-peak hours, maintaining continuous operation and ensuring consistent processing of waste materials. This predictive maintenance capability, supported by IoT data, reduces operational disruptions and extends the lifespan of sorting machinery, providing significant cost savings over the long term. Additionally, real-time monitoring helps facility managers make data-driven decisions that improve operational efficiency and reduce energy consumption, aligning MRF operations with sustainability goals. One of the most promising aspects of AI and IoT in MRFs is their potential to optimize waste recovery rates by adapting to shifts in waste composition. For instance, as packaging materials evolve, MRFs must be able to handle new materials that may not have been common in the waste stream before. AI-powered sorting systems are well-suited for this challenge, as they can be programmed to recognize and separate new materials without requiring significant manual reconfiguration. This adaptability ensures that MRFs can keep up with changing consumer trends and waste patterns, maximizing the amount of recyclable material they recover. In some facilities, AI algorithms are even used to forecast future waste trends based on historical data, helping MRFs prepare for anticipated changes in waste composition and maintain high recovery rates. The operational data generated by AI and IoT systems can also be shared with local governments, waste collection agencies, and recycling companies, thereby providing a comprehensive overview of the waste management ecosystem. By sharing information about waste composition and volume, stakeholders can make informed decisions about resource allocation, recycling program development, and public awareness campaigns aimed at reducing waste generation. For example, if data from MRFs show a high volume of single-use plastics in the waste stream, local governments might consider implementing policies to encourage the use of reusable alternatives. This data-driven approach invites more effective and targeted interventions that address specific waste challenges within an urban community. Moreover, the automation of sorting processes enabled by AI reduces the need for manual labor in MRFs, addressing both labor shortages and health risks associated with waste handling. Sorting waste manually is labor-intensive and exposes workers to hazardous materials and unsanitary conditions. By automating sorting tasks, AI-powered systems not only increase productivity but also improve safety for workers by minimizing their direct contact with waste. This benefit is particularly relevant given the risk of future pandemics comparable to COVID-19, which has highlighted the importance of minimizing human exposure in high-risk environments. Robotics and AI systems provide a viable solution to these challenges, ensuring that MRFs can operate safely and efficiently with a reduced reliance on human labor. Furthermore, the increased sorting accuracy enabled by AI means higher-quality recycled materials, which are then more valuable in the market. Recycling facilities that receive well-sorted materials from MRFs can process them more efficiently and at a lower cost, making recycling more economically viable. High-quality recycled materials can then be sold to manufacturers at a premium, creating a positive revenue stream for waste management systems and encouraging further investment in recycling infrastructure. In this way, AI-driven sorting not only supports environmental goals but also contributes to the financial sustainability of recycling programs. The role of AI in MRFs further extends to sustainability by promoting the concept of a circular economy, where materials are continuously reused rather than disposed of after a single use. By ensuring that recyclable materials are efficiently separated from the waste stream and returned to the manufacturing process, AI-powered sorting systems help reduce the demand for virgin resources, conserve energy, and decrease greenhouse gas emissions associated with the production of new materials. For example, recycling aluminum saves 95 % of the energy required to produce it from raw materials, while recycling plastic can reduce emissions and resource use compared to producing new plastic from petrochemicals. AI and IoT technologies in MRFs thus play a crucial role in closing the loop on resource use and supporting the transition to a more sustainable economy. Looking ahead, the potential for AI and IoT to further transform MRFs is vast. Future advancements in ML algorithms, combined with improvements in robotics and sensor technology, are likely to fine-tune sorting precision. Researchers are exploring the use of neural networks to enhance AI's ability to recognize and classify complex waste items, as well as the potential for AI systems to sort multi-material objects that contain both recyclable and non-recyclable components. These advancements will enable MRFs to handle an even wider range of waste materials, improving recovery rates and further reducing the amount of waste that ends up in landfills.

4. Enhanced recycling through smart technologies

Blockchain technology, when integrated with IoT sensors, has the potential to revolutionize the waste management industry by creating a transparent and traceable journey for recyclable materials from collection to final processing [10,19]. This system allows each item of recyclable waste to be tracked and verified, ensuring that it reaches the proper processing facility and is ultimately recycled rather than diverted to landfills. For consumers, such transparency builds confidence, as they can verify that their recycling efforts have tangible outcomes. Waste management companies also benefit from increased public trust and accountability, reinforcing their commitment to sustainable practices. By enabling a secure, immutable record of each item's journey through the recycling process, blockchain enhances traceability, reduces the risk of diversion, and ensures compliance with recycling regulations and goals. In addition to traceability, AI-driven technology accurately identifies and sorts specific recyclable materials with unmatched accuracy. Modern AI systems can distinguish between different grades of plastics as well as various types of glass and metals, allowing for precise categorization and separation. Meticulous sorting is vital for the recycling industry, as it directly impacts the quality of the materials recovered. Higher-quality sorting produces materials that are purer and more suitable for high-value applications in the market, creating a cycle where recycled materials are increasingly able to meet industry standards and replace virgin materials in production. Consequently, the demand for high-quality recyclables grows, supporting the broader adoption of recycled content in manufacturing and reducing the need for resource extraction.

To foster greater public participation in recycling initiatives, many cities and companies are adopting data-driven approaches, leveraging AI-powered platforms and mobile applications to track individual contributions to recycling programs. By providing real-time feedback on recycling habits and offering rewards or discounts for active participation, these platforms encourage responsible waste disposal and make recycling a more engaging and rewarding experience. Citizens can earn incentives, which might include discounts on local services or even cash rewards, by meeting recycling benchmarks or correctly sorting waste at designated drop-off points. This gamified approach nurtures a culture of sustainability by aligning economic incentives with environmentally friendly behavior, thus promoting a circular economy. The data gathered through these platforms also enables cities and companies to better understand recycling patterns, optimize waste collection schedules, and target areas where additional recycling education may be needed and/or beneficial.

5. Integration of waste data analytics for decision-Making

Data collection and analytics in urban waste management represent a transformative approach that allows municipalities to gather detailed information on waste patterns, disposal habits, and collection

efficiencies [43,46]. IoT devices embedded in waste bins and collection vehicles continuously gather data on waste volumes, bin usage, and disposal frequency. This influx of real-time information provides city planners with an extensive dataset that, when analyzed, reveals trends and patterns in waste disposal across different neighborhoods [12,46]. By understanding these patterns, city officials can make informed decisions on the optimal allocation of resources, ensuring the efficiency of waste collection while avoiding the over-servicing of certain areas. For example, high-traffic or high-density areas might require more frequent waste collection, while residential areas with lower waste volumes can benefit from a less frequent schedule. This data-driven insight enables municipalities to create a tailored waste management approach that not only meets the unique needs of each community, but also reduces collection costs and enhances service delivery across the city.

AI-based predictive models take this data analysis to a new level by helping urban areas anticipate waste generation patterns based on various factors, including population density, waste type, and seasonal variations. These algorithms can integrate data on past waste generation and demographic factors, creating accurate predictions for future waste levels [29]. For instance, during holiday seasons or city-wide events, waste production often spikes. Predictive models can help cities prepare accordingly by adjusting waste collection schedules and deploying additional resources in advance. This foresight reduces instances of overflowing bins and litter while also minimizing the risk of increased pollution from uncollected waste. Furthermore, by predicting which neighborhoods will likely experience more waste generation due to population growth or commercial expansion, municipalities can proactively plan infrastructure improvements, such as adding more bins or increasing collection frequency in those areas. By enabling a dynamic response to waste generation trends, predictive modeling strengthens waste management systems, making them more resilient to changing urban demands.

From the above, it is clear that any data collected from IoT-enabled bins and collection vehicles can significantly improve operational efficiency. The data also serve as a robust tool for municipal decision-making support, allowing city planners and policymakers to make more informed choices when designing waste management policies, infrastructure investments, and public awareness programs. For instance, if the data reveal that a particular area generates a significant amount of plastic waste, the city could introduce targeted recycling initiatives or educational campaigns focused on reducing plastic usage in that neighborhood. These targeted efforts help optimize public engagement and ensure that resources are allocated to the areas where they can have the greatest impact. Similarly, the data can support infrastructure planning, such as deciding where to locate new recycling facilities or composting sites based on waste generation hotspots. The aggregated insights from the collected data will provide a foundation for developing ongoing waste management strategies that align with broader environmental goals, such as reducing landfill dependency or increasing recycling rates. This strategic planning capability, supported by data-driven insights, enables municipalities to create sustainable waste management frameworks that benefit both the environment and urban residents.

In addition to enhancing operational efficiencies and supporting informed policy-making, IoT-enabled data collection allows cities to conduct in-depth analyses of waste management costs and environmental impacts [3,53], waste diversion rates, landfill volumes, and recycling efficiency. Based on these analyses, a more comprehensive picture of a city's sustainability efforts can be discerned, thus guiding individual cities to set measurable and realistic targets for themselves, such as reducing landfill waste by a certain percentage each year or increasing recycling rates. With a clear view of their current performance and progress, cities can align their waste management efforts with regional or national environmental targets, fostering accountability and measurable improvements over time. Beyond operational and environmental benefits, data-driven waste management offers significant advantages for public engagement and awareness. By sharing data insights with the public, municipalities can increase transparency and encourage community participation in waste reduction efforts. Many cities, for instance, have introduced apps or online platforms where residents can see real-time information on waste collection schedules, recycling rates, and local environmental initiatives. Such platforms foster a sense of shared responsibility, as residents become more aware of their individual impact on the city's waste levels and are encouraged to adopt sustainable habits. Furthermore, municipalities can use data insights to organize targeted public awareness campaigns. For example, if the data indicate that some neighborhoods have a lower recycling rate, the city can launch educational workshops or provide informational materials on proper recycling practices targeting those areas. This targeted approach to public education ensures that resources are allocated effectively, reaching the communities that most need guidance in sustainable waste disposal. Over time, collaboration between city planners and residents can help build a culture of sustainability, where individuals feel empowered to contribute voluntarily to waste reduction and environmental stewardship.

As IoT devices and AI models continue to evolve, their integration in MSW management systems will allow for increasingly sophisticated analyses and applications [53]. One emerging application is the integration of smart sensors that can measure not only waste volumes but also specific environmental factors, such as air quality or temperature, around waste collection sites. By analyzing correlations between waste disposal patterns and environmental conditions, cities can identify and mitigate potential health risks associated with waste accumulation, particularly in densely populated urban areas. As these technologies advance, the accuracy and granularity of data collection will improve, allowing cities to pinpoint waste sources more precisely and address specific behaviors contributing to waste challenges. For instance, if certain types of businesses are shown to generate disproportionate amounts of waste, cities can introduce policies or incentives to encourage waste reduction within that sector. Such a targeted intervention strategy ensures that waste management efforts are efficient and directly address the sources of waste generation, maximizing the impact of sustainability initiatives.

Additionally, the ability to monitor waste flow in real-time across neighborhoods offers opportunities for adaptive resource management. If a sudden surge in waste is detected in a particular area, such as after a major event or festival, the city can deploy additional resources or adjust collection schedules in real-time to prevent overflowing bins and litter. High levels of responsiveness enhance public cleanliness and reduce the likelihood of environmental damage from uncollected waste. The predictive capabilities of AI, combined with real-time IoT data, enable waste management systems to operate dynamically, adjusting to immediate needs and minimizing waste-related disruptions in the urban landscape. Another critical advantage of data-driven waste management is its role in supporting the circular economy [38]. By identifying waste composition patterns, cities can streamline recycling processes and improve resource recovery rates. For example, if the relevant data show a high prevalence of electronic waste in certain areas, municipalities can introduce specialized e-waste collection drives or recycling programs, ensuring that valuable materials like metals and rare earth elements are recovered and reused [17]. This approach not only reduces landfill dependency but also supports local industries by providing a reliable source of recycled materials. Over time, as more resources are recovered from the waste stream, cities can move closer to achieving circular economy goals, where materials are continually repurposed rather than discarded, reducing the environmental and economic costs of resource extraction. IoT-enabled data collection and AI-driven analysis are also instrumental in tracking and managing hazardous waste materials. In many urban areas, improper disposal of hazardous materials, such as batteries, chemicals, and electronic waste, poses significant environmental and health risks [13,18]. By using sensors to detect and classify hazardous materials, MSW management systems can ensure that these

materials are handled appropriately and do not contaminate the general waste stream. This capability is particularly valuable in industrial areas, where hazardous waste generation is higher, as it enables cities to enforce proper disposal practices and protect public health. The data gathered from hazardous waste tracking can inform policy decisions, such as introducing stricter regulations or increasing disposal fees for high-risk materials. If they address hazardous waste at its source, cities can create safer and cleaner urban environments for their residents.

The combined power of IoT and AI in waste management throws open the door to collaborative partnerships between municipalities and the private sector. Many businesses, especially those in manufacturing and retail, produce significant amounts of waste, making them key stakeholders in municipal waste management strategies. By sharing data insights with private companies, cities can encourage businesses to adopt more sustainable waste disposal practices and even participate in waste reduction initiatives, such as recycling programs or waste-to-energy projects. Companies can use municipal data to optimize their own waste management practices, reducing waste production and lowering costs. These partnerships benefit both cities and businesses by creating a coordinated approach to waste management that aligns with sustainability goals. Over time, as more companies participate in these initiatives, the city's overall waste levels decrease, further enhancing the effectiveness of municipal waste management efforts. Ultimately, the adoption of data collection and analytics, powered by IoT and AI, represents a paradigm shift in urban waste management. The real-time insights, predictive capabilities, and decision-making support offered by these technologies empower cities to create efficient, responsive, and sustainable waste management systems that meet the demands of their unique environments. Furthermore, given the evolution of IoT and AI, these technologies have the potential for even more innovative applications in waste management. As cities tackle growing waste challenges, data-driven strategies are essential for resource allocation, public engagement, and long-term environmental goals. Another key aim is to engage residents in sustainable practices in order to boost the success rate of recycling programs. To accomplish this, cities can use real-time data to create educational campaigns, provide recycling incentives, and promote waste sorting and composting. Interactive platforms and social media can also spread awareness, while gamification and community events can incentivize participation. Through fostering a sense of responsibility and offering rewards for sustainable actions, cities can encourage active involvement in waste reduction. This strategy not only enhances operational efficiency but also builds a cleaner, healthier, and more sustainable urban environment that succeeds in setting a new waste management standard.

6. Challenges and limitations

Despite their clear benefits, AI and IoT systems can be challenging to implement in MSW management, particularly with regard to high initial costs, privacy and security concerns, and technical limitations [4,14]. The financial burden associated with setting up these advanced systems may seem daunting to many municipalities, especially those with constrained budgets, as the initial investment required for hardware, software, installation, and integration is substantial. For example, deploying a network of IoT sensors across an entire city requires not only the cost of the sensors themselves but also investment in supporting infrastructure, such as data management systems and cloud services for real-time analytics. Moreover, cities must consider the costs associated with training personnel to operate and maintain the installed systems. Although long-term savings in operational efficiencies and reduced waste disposal costs can offset these initial expenditures, the upfront financial outlay often serves as a barrier for municipalities. This financial hesitance can prevent cities from adopting innovative waste management solutions that could ultimately lead to more sustainable practices, overall cheaper costs, and better environmental outcomes.

Along with financial concerns, privacy and security issues represent another critical challenge in the deployment of IoT devices for MSW management. These devices collect detailed data on residents' waste disposal habits, which can inadvertently infringe on their personal privacy. Some people may feel uncomfortable with their disposal behaviors being monitored, leading to potential pushback against the implementation. Moreover, the collection of sensitive data necessitates the establishment of robust data privacy policies to safeguard residents' information. Cities must be transparent about the type of data being collected, how the data will be used, and who will have access. Without strong protections in place, there is a risk of data misuse, whether through breaches or unauthorized sharing with third parties. To build trust with the public, municipalities need to engage in proactive communication regarding their data handling practices and ensure compliance with existing privacy regulations. Establishing a framework for accountability and transparency is essential not only for addressing residents' concerns but also for fostering a culture of trust that will facilitate broader acceptance of IoT and AI technologies in MSW management. Technical limitations present yet another layer of complexity in the implementation of AI and IoT systems. Although these technologies are designed to improve operational efficiencies and accuracy in waste management processes, their performance can sometimes be inconsistent. For instance, inaccuracies in material identification can lead to improper sorting of recyclables, undermining the very purpose of implementing the advanced systems. Such errors can occur due to a variety of factors, including limitations in the algorithms used for machine learning or environmental conditions that obscure sensor readings. Furthermore, technical malfunctions such as sensor failures or connectivity issues can disrupt operations, leading to delays in waste collection and increased risks of overflow or contamination.

To address these challenges, cities must invest not only in the initial setup of the technologies but also in ongoing maintenance and support. This includes hiring skilled technicians who can troubleshoot issues, perform regular maintenance, and ensure that systems operate reliably over time. Additionally, continuous training and development for personnel involved in managing these technologies are crucial to keeping pace with advancements and ensuring optimal performance. By recognizing and addressing these technical limitations, municipalities can enhance the reliability and effectiveness of AI and IoT systems in waste management, ultimately leading to better service delivery and improved sustainability outcomes.

7. Future directions and research opportunities

The integration of AI and IoT technologies in waste management represents a significant shift towards more sustainable practices, particularly in urban environments where waste generation is high and increasing. This transformation focuses on improving collection and sorting efficiency, with an emphasis on waste prevention at the source. By leveraging predictive analytics, smart systems can promote sustainable consumption patterns, reduce excess waste generation, and facilitate a transition to a more circular economy. Moreover, advancements in robotics for sorting waste and the necessity for supportive policies and regulatory frameworks will be crucial in realizing the fascinating and ever-evolving potential of these technologies. AI-driven waste prevention is an emerging focus that seeks to address waste generation before it occurs. By utilizing predictive analytics, municipalities can gain insights into their unique consumption patterns and identify areas where waste can be minimized. For example, grocery stores can leverage AI algorithms to analyze sales data, forecast demand, and adjust inventory levels accordingly. This proactive approach helps reduce food waste while also minimizing excess inventory, leading to cost savings for retailers. Similarly, manufacturers can use AI to optimize production processes and reduce material waste during manufacturing. By integrating waste prevention strategies into their operational frameworks, businesses can contribute to a more sustainable ecosystem and enhance their bottom line. Unilever is a prime example of a prominent

multinational consumer goods company that has adopted AI-driven analytics to reduce waste across its supply chain. Using ML algorithms to analyze consumer purchasing patterns, the company can predict which products are likely to generate excess waste and adjust production schedules accordingly. This data-driven approach enables Unilever to minimize overproduction and associated waste, reinforcing its commitment to sustainable practices. Moreover, AI can empower consumers by providing personalized insights into their consumption habits, helping them make more informed choices that reduce waste at home. Mobile applications can offer suggestions for meal planning based on users' purchasing behaviors, encouraging them to buy only what they need and ultimately minimizing food waste. Furthermore, advanced robotics in waste sorting are revolutionizing the recycling landscape, making it feasible for cities to process waste at unprecedented scales. Robotic automation is increasingly being integrated into MRFs to enhance sorting accuracy and efficiency. AI-powered robots can quickly and precisely identify different materials based on shape, size, and composition, significantly increasing recycling rates compared to human manual labor, and helping cities to achieve ambitious zero-waste goals.

A notable example of this advancement is the implementation of robotic sorting systems by AMP Robotics, a company specializing in AI-driven waste sorting technologies. Their robotic systems use computer vision and machine learning to recognize various types of recyclable materials, reducing contamination rates and increasing the quality of recovered materials. This not only maximizes recycling outputs but also minimizes the amount of waste sent to landfills. As cities strive to meet zero-waste targets, advanced robotics will play an essential role in streamlining sorting operations and facilitating the transition to circular economy models. In addition to technological advancements, effective policy and regulatory support are paramount for the successful implementation of AI and IoT technologies in MSW management. Governments can create a favorable environment for innovation by establishing clear standards for smart waste management systems, providing guidelines and frameworks, ensuring consistency and efficiency across cities, and fostering collaboration between public and private sectors. For instance, the city of San Francisco has implemented policies to promote smart waste management technologies, providing generous funding opportunities for initiatives that focus on municipal waste reduction and recycling enhancements. Public-private partnerships can also be instrumental in facilitating the adoption of advanced waste management solutions. By collaborating with technology providers, municipalities can leverage expertise and resources to develop innovative systems that improve waste collection, sorting, and recycling processes. An example is the partnership between the city of Los Angeles and the waste management company Republic Services, which has implemented smart waste management solutions to enhance operational efficiency. Through the use of IoT sensors, the city can monitor the fullness of bins in real-time, optimizing collection routes and reducing costs associated with fuel consumption and labor. This collaboration exemplifies how public-private partnerships can drive innovation and improve waste management outcomes. Moreover, funding opportunities are crucial for cities looking to invest in AI and IoT technologies. Governments can allocate resources to support the development and deployment of smart waste management solutions, enabling municipalities to overcome financial barriers associated with initial investments. By establishing grants or incentives for cities to adopt innovative technologies, authorities can encourage widespread adoption and accelerate the transition toward sustainable waste management practices. For instance, the European Union has provided funding through programs like Horizonto support projects focused on developing smart waste management solutions across member states. These initiatives not only provide financial assistance but also encourage collaboration among stakeholders by rewarding participation and promoting knowledge-sharing and best practices in waste management. In addition to financial support, public awareness campaigns are essential for ensuring the success of implementing AI and IoT technologies into waste management. Educating citizens about the benefits of smart waste systems and how they contribute to sustainability can help encourage community participation in recycling initiatives. Cities can leverage social media platforms and community events to raise awareness about proper waste disposal practices, the importance of recycling, and the role of technology in enhancing waste management outcomes. Engaging citizens in these conversations creates a sense of ownership and accountability, ultimately leading to more responsible consumption and disposal behaviors. As cities continue to grapple with the challenges of waste management, the integration of AI and IoT technologies offers a promising pathway to enhanced sustainability and efficiency. However, specific actionable strategies must be applied to ensure collaboration between governments, technology providers, and the public. One effective strategy is establishing public-private partnerships (PPPs), where governments can provide incentives, subsidies, or tax breaks to companies that invest in innovative waste management technologies. This could include grants for pilot projects focused on smart bins, AI-powered sorting, or IoT-based route optimization, enabling private companies to scale these solutions while ensuring alignment with public goals. In parallel, local governments should initiate public awareness campaigns focused on the benefits of smart waste management, emphasizing how residents and businesses can appreciably contribute to sustainability efforts. This can include educational programs, community workshops, and digital platforms to track and reward waste reduction and recycling efforts. These programs could integrate AI-driven apps where citizens can monitor their waste management habits, access recycling tips, and receive incentives and rewards for responsible consumption and disposal. Furthermore, municipalities can adopt regulatory frameworks that set clear guidelines for waste reduction and recycling, coupled with incentives for early adopters of and investors in AI and IoT-based waste management solutions. This can be achieved through progressive laws that reward high-performing cities or companies that implement smart waste solutions effectively, while imposing penalties for those who continue to rely on outdated systems that harm the environment. In so doing, governments can ensure that policies are not only supportive but also enforceable, encouraging consistent progress toward sustainable waste management goals. Incorporating these strategies into policy frameworks will help to create a clear and actionable pathway for collaboration that will ensure the integration of AI and IoT technologies into MSW management systems. Through close coordination between government bodies, private sector innovators, and the public, municipalities can employ these technologies to reduce their waste generation and increase recycling rates, ultimately building a more sustainable and circular urban economy.

8. Conclusion

The integration of AI and IoT technologies into urban solid waste management represents a transformative shift, offering promising solutions to address the growing challenges of waste disposal and recycling in rapidly expanding cities. As municipal populations increase and consumption patterns evolve, cities are facing significant waste management difficulties that jeopardize environmental sustainability and public health. AI and IoT technologies provide innovative solutions that not only enhance operational efficiency but also create a foundation for more sustainable urban ecosystems. Through the optimization of collection, sorting, and recycling processes, these technologies play a pivotal role in minimizing urban environmental footprints, supporting a circular economy, and helping cities meet their sustainability goals.

The benefits of AI and IoT in waste management span the various stages of the waste lifecycle. AI-driven predictive analytics further strengthen waste management systems by enabling cities to forecast waste generation based on factors such as population density, seasonal variations, and community events. Machine-learning algorithms allow municipalities to optimize waste collection schedules, ensuring that bins

are emptied before the overflow, which reduces littering and potential public health risks. This proactive approach also enhances user experience for residents and fosters a culture of sustainability in communities.

Moreover, advanced robotics integration in sorting facilities is revolutionizing how recyclable materials are processed. AI-powered sorting systems utilize computer vision technology to precisely identify and classify waste, significantly reducing manual labor and increasing sorting accuracy. Such automation is essential for improving recycling rates and advancing the goal of a zero-waste society. As cities strive for ambitious recycling targets, robotic sorting systems greatly enhance the efficiency and effectiveness of MRFs.

Smart waste management systems employ IoT devices to collect real-time data on waste volumes, disposal frequencies, and bin usage, enabling municipalities to analyze waste flow patterns and utilize resources more effectively. Through IoT technology, cities can identify high-waste areas and adjust collection schedules accordingly. This approach enhances operational efficiency while also reducing costs associated with fuel consumption and labor, making waste management more economically viable.

As AI and IoT technologies continue to evolve, their potential to revolutionize waste management will only expand. However, for cities to fully capitalize on these technologies, strong collaboration between governments, technology providers, and the public is crucial. Policymakers must create a regulatory environment that facilitates the adoption of smart waste management solutions by establishing standards for data privacy, ensuring equitable access to technology, and providing funding opportunities for municipalities to implement these innovations. Public-private partnerships are also critical, as they combine resources and expertise to drive the development and deployment of cutting-edge waste management solutions.

Because public engagement is essential for the success of AI and IoT technologies in waste management, educating citizens about the benefits of smart MSW management systems and fostering community participation are vital. Social media, local outreach, and community events can be used by cities to raise awareness about proper waste disposal and recycling practices. When citizens actively participate in these initiatives, they take on greater responsibility for sustainable habits and contribute to higher recycling rates.

Furthermore, data collected from AI and IoT systems provides valuable insights that can guide waste management policies and strategies. By analyzing data on waste generation patterns, cities can design more efficient infrastructure, introduce new recycling programs, and implement targeted public awareness campaigns, ensuring that policies are grounded in evidence for maximum impact.

Despite the advantages of AI and IoT, some challenges remain, including high initial costs, privacy concerns, and technical limitations. To overcome these obstacles, policymakers can offer financial support, grants, and incentives to assist cities in investing in these technologies. Governments should also establish robust data privacy policies to protect their citizens' personal information and ensure that IoT devices are used solely for waste management purposes. Addressing these concerns can build trust and encourage greater public participation.

Technical limitations involving inaccuracies in material identification or system malfunctions must also be addressed. This can be accomplished through continuous system maintenance, skilled technicians, and regular training. Cities must invest in ensuring the reliability and effectiveness of these technologies to fully realize their potential. Policymakers should also establish technical standards to ensure that AI and IoT systems function optimally across different urban contexts.

If the vast potential of AI and IoT in urban waste management is to be realized, collaboration and innovation must be at the forefront of implementation efforts. Strategic investments in advanced technologies, coupled with public awareness campaigns and a commitment to fostering a culture of sustainability, will create a resilient circular urban system that minimizes environmental harm, conserves resources, and contributes to a greener future. Through this integrated approach, cities will not only address the immediate challenges of waste management but also contribute to broader goals such as climate change mitigation and environmental justice, ultimately fostering more livable urban environments. By embracing these strategies, communities can pave the way to a cleaner and more sustainable future for generations to come.

CRediT authorship contribution statement

Abderrahim Lakhouit: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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